



Planning Formula One race strategies using discrete-event simulation

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A discrete-event simulation model that imitates most on-track events, including car failures, passing manoeuvres and pit stops during a Formula One race, is presented. The model is intended for use by a specific team. It will enable decision-makers to plan and evaluate their race strategy, consequently providing them with a possible competitive advantage. The simulation modelling approach presented in this paper captures the mechanical complexities and physical interactions of a race car with its environment through a time-based approach. Model verification and validation are demonstrated using three races from the 2005 season. The application of the model is illustrated by evaluating the race strategies employed by a specific team during these three races.

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Introduction

Large sums of money are invested annually in professional sports and sponsors reward successful individuals and teams to a much greater extent than less successful ones. Sponsors also pressure individuals and teams to succeed, leaving them little room for error.

Operational research has been widely applied to sports (Gerchak, 1994; Hurley, 2006), while simulation has found application in its various forms over a wide spectrum. Simulation in sport has, for example, been used to analyse and improve human performance in specific sporting activities like cycling (Vilimek, 2006), jumping (Domire and Challis, 2004) and skating (Allinger and van den Bogert, 1997). It is also used in sports equipment design (Brannigan and Adali, 1981; Hendry and Hubbard, 2000; Corazza and Cobelli, 2005), tournament design (O'Donoghue, 2005), and predicting the outcomes of sports tournaments (Koning *et al*, 2003; Barnett and Clarke, 2005). In this paper, the development and application of a discrete-event computer simulation model in Formula One car racing is discussed.

In most sporting competitions, the success of a team depends on the abilities of the participating sportsmen and women, the equipment used (where applicable), the management of the team and the game strategy planned. Formula One car racing is a multi-million dollar sport involving racing cars that are designed, manufactured and managed using cutting-edge technology (F1Technical, accessed 2008) and is followed by millions of people around the world each year

(Premiere, accessed 2008). A single wrong decision before or during a race could easily lead to failure. As a result, teams constantly seek aids and means that could provide them with a competitive advantage. These aids and means must ensure the optimization of resources in the quest to achieve the best race results possible.

The main objective of this research was to develop a simulation model capable of imitating the essentials of race car movement around a race track, as well as the major events during a Formula One race. The simulation model presented in this paper can assist a racing team with the planning and evaluation of its race strategies in order to improve the team's final placement in a race. Simulation is a valuable decision support tool, and in this study it specifically allowed for evaluation of scenarios with stochastic elements.

In the following sections, a background to Formula One racing is given to provide the reader with a better understanding of the world's leading motor sport category. This is followed by a discussion of the simulation concept and the modelling assumptions, after which the nature and requirements of the input data for the model, as well as the verification and validation of the simulation model are explained. Finally, the results are presented and conclusions are drawn.

Background to Formula One racing

Formula One (F1) racing originated in 1947 from European Grand Prix racing, and the first FIA (*Fédération Internationale de l'Automobile*, or International Automobile Federation, see FIA, accessed 2008) world championship race was held in 1950 (Hayhoe and Holland, 2006). The term Formula One refers to a set of technical regulations for single-seater

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racing cars that are published annually by FIA, the governing body of motor sport. An F1 racing car is built to these regulations and competes against other F1 cars in an F1 race, also known as *Grand Prix*. This title is reserved, by FIA, for events counting towards the F1 world championship (Formula One database, accessed 2008). The past few decades have seen F1 racing tour the world during the annual racing season with racing events held in countries such as Australia, Germany, the United Kingdom and the USA. The 2005 season comprised 19 races and attracted a cumulative television audience of approximately 800 million (Premiere, accessed 2008).

Over the years, F1 racing has seen many changes as technology improved, rules and regulations changed, and more and more money was invested by sponsors. The principle, however, has remained the same: specially built cars race against each other on a race track for a predetermined number of laps, and after each race, the more successful drivers are awarded points depending on their final positions, with the associated teams receiving the same number of points. At the end of the season, the driver with the most points receives the Drivers' World Championship title, while the teams compete for the Constructors' World Championship title. The top driver and team at the end of the season attract much prestige and often substantial future sponsorships. Currently a team (constructor) consists of two drivers (two cars), a large number of supporting staff, equipment and a management team.

The day before a race, all participating drivers and their associated cars are required to participate in a qualifying session that determines the starting order of a race. This is achieved by sorting the drivers' qualifying times in ascending order, with the faster cars at the front of a two-row starting arrangement. During a race, a driver may temporarily suspend participation and visit his support team in a designated area called the pits; this activity is known as a pit stop. During a pit stop the car may be refuelled, and may have its tyres changed, damaged body parts replaced and/or some elements of its setup changed. In doing so, the driver loses time with respect to the other competitors who continue the race, and after the pit stop he then tries to catch up on the track. Cars sometimes retire from a race due to technical problems or accidents.

The simulation model design

When imitating the behaviour of racing cars, many characteristics of the cars should be considered, including air resistance, rolling resistance, engine power, car mass, torque range and tyre material (tyre compounds). Milliken and Milliken (1995) provide a comprehensive source of race car vehicle dynamics that considers the various properties of a race car. Environmental conditions also play a significant role in the performance of race cars, for example, the surface of a hot and dry race track has a different grip ability than a damp or wet track and therefore requires different tyres. Wind may also affect the aerodynamic performance of a car reducing the performance level.

In-race events also influence the cars, notably when accidents occur during the usual mixing of cars at the start of a race. As mentioned earlier, the cars are lined up at the start in a two-row configuration called the starting grid, and as soon as the race commences, drivers lower down the grid immediately try to pass those higher up in the grid. The order in which the drivers start the race changes very quickly due to the dissimilarity in driving skill, reaction time and the launch control capability of a car. Other in-race events include the passing of opponents, and pit stops.

Currently, many racing simulations exist, particularly in the entertainment industry. The developers attempt to make these as realistic as possible using the expertise of automotive engineers and race car drivers, excellent graphics and supporting equipment like racing chairs. The physics of the race car are also included in these simulation games to different degrees. Perhaps the most advanced product currently is Forza Motorsport 2 that runs on Xbox 360 (Forza Motorsport 2 Product Info, accessed 2008). In these games, one or more players participate in a race, and the outcome mainly depends on the skill and the experience of the player(s).

Our simulation approach is different, as we allow the cars to interact according to conditions and probabilities, with no human intervention during the simulated race. Our time-based approach, which will be discussed later, assumes that all effects result in a net time that a race car takes to travel from one point to another on the race track, and thus allows simpler, repeatable simulations.

In this research, the *dynamics of a race* are modelled to aid the planning and evaluation of race strategies. The effects cars have on each other while on a given section of track must be accounted for, for example their ability to pass one another. The simulation world view of Shannon (1975, p. 109) was used as a modelling paradigm. It is formulated as follows: *entities* having *attributes* interact with resources or *activities* under certain *conditions*, creating *events* that change the *state of the system*. In modelling a race, the racing cars are the entities that have attributes, for example, the amount of fuel on board. These entities interact with activities, for example the need to perform a pit stop or attempting a passing manoeuvre under certain conditions. An event in the context of this paradigm is a pit stop, or a car retiring from a race. The system is the collection of entities (cars) and the race track definition (distance, typical lap times), while the state of the system is a collection of state variables. These variables include the instantaneous positions of all cars, the number of laps completed by each car and the number of cars racing at any point in time. The model was implemented in Arena[®] 9.0 (Rockwell Automation, accessed 2008), as it supports the paradigm described.

A time-adjustment approach

With the above in mind, a time-adjustment approach was developed to simulate a Formula One race. This approach can

be explained as follows: The potential of drivers and their associated cars differ from one another, in some cases significantly. These dissimilarities are caused by several factors, such as driver skill and experience, car design and setup, and the size of a team's budget. Essentially, the driver potential translates to how fast a driver is able to complete a single lap. As a result, the potential of each driver is contained in an attribute called *base lap time*, which is the fastest time a driver achieves during the practice and qualifying sessions leading up to a race. This potential is then assumed to be the maximum capability of the car, which means that if the car had practically no fuel on board, the driver had completed a flawless lap and there had been no impeding traffic, one would expect a lap time to equal the base lap time.

Prior to simulating a race, the race track is divided into a number of sectors. Currently, a race track is divided into a nominal 40 sectors of approximately 150 m each. Increasing the number of sectors may increase the accuracy of the model,

albeit at the price of computational time. Once the sectors have been created, the race track characteristics corresponding to each sector are assigned, for example, whether the sector permits passing manoeuvres or not, the relative proportion of time spent in the sector, and the proportion of fuel consumed in the sector. The time a car spends in each sector and the fuel consumed in each sector may be obtained from telemetry data, from which the model calculates the simulated time each car spends in all the sectors, that is the proportion of the lap spent in a sector is multiplied by the individual base lap time of each car. The fuel consumed in each sector is determined in a similar manner.

The simulation model allows each entity (race car), to move from one sector to another, one after the other. The time it takes to complete a sector is calculated by adding time adjustments to the proportion of the base lap time, assigned to the sector in question, for each entity as it enters a sector. These time adjustments are based on various attributes of both race

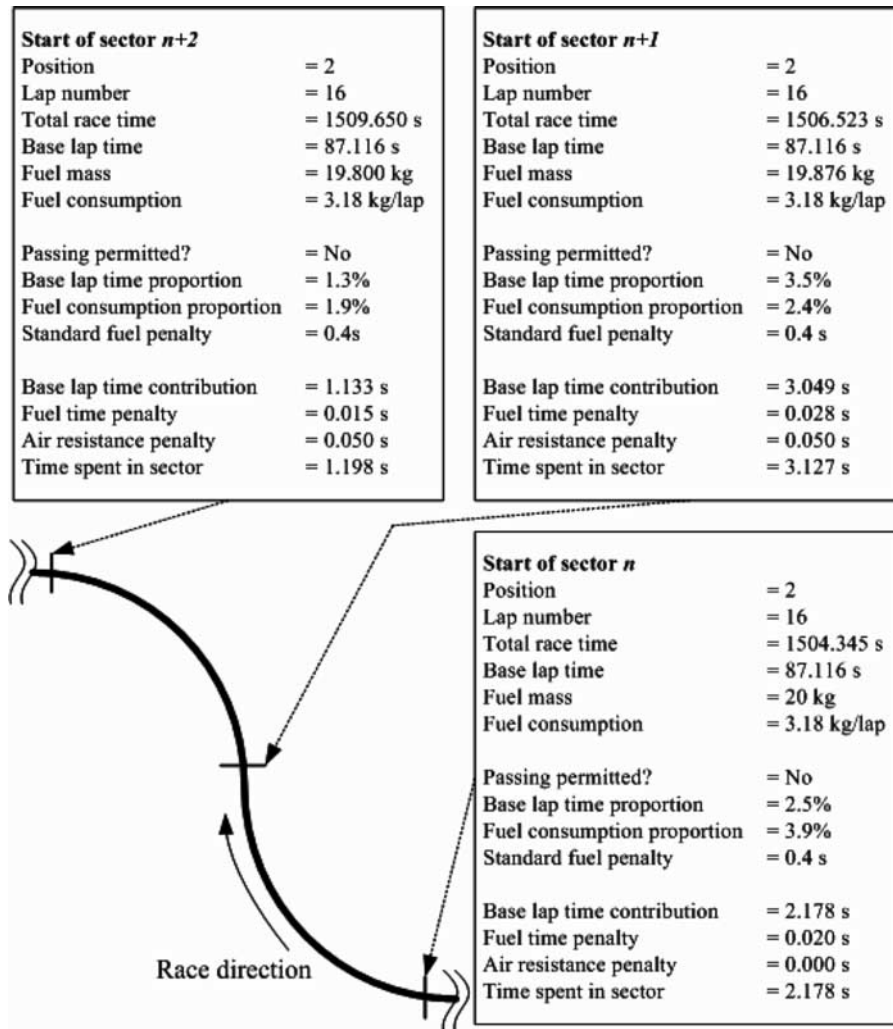


Figure 1 An illustration of the time-adjustment principle for a car moving from sector n to sector $n+2$.

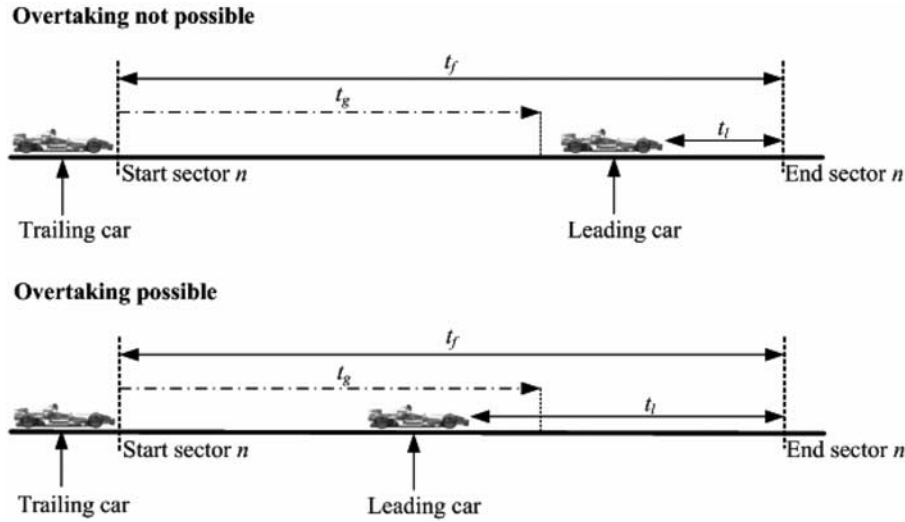


Figure 2 An illustration of the passing manoeuvre concept.

track and race car, for example, the current position occupied by the driver, the amount of fuel on board the associated car at that point in time, the fuel consumption of the associated car, the nature of the traffic created by opposition drivers and the characteristics of the sector a driver occupies (straight, or a corner). Some time adjustments favour the car; for example, the fuel consumed in a sector makes the car lighter and as a result faster. On the other hand, when an opponent is closely followed, the second car experiences turbulence, thus slowing the car down. The above-mentioned principles are illustrated in Figure 1.

As an example, assume a car occupying the k th position upon entering the n th sector on the j th lap has a fuel load of 20 kg. Also assume that 0.4 s are added to the base lap time for every 10 kg of fuel on board the car and that a race track analysis indicates the lap proportion spent in sector n to be 2.5%. The *fuel time penalty* is the first in a number of time adjustments that need to be made, and is calculated as follows:

$$\begin{aligned}
 T_{FP} &= L_P \times f_{MF} \times \frac{m(t)}{m_N} \\
 &= 0.025 \times 0.4 \text{ s} \times \frac{20 \text{ kg}}{10 \text{ kg}} \\
 &= 0.01 \text{ s}
 \end{aligned} \tag{1}$$

where T_{FP} denotes the fuel time penalty [s], L_P denotes the lap proportion spent in sector n , f_{MF} denotes the time penalty for every 10 kg fuel carried per lap (s), $m(t)$ denotes the fuel mass on board at time t (kg) and m_N denotes the standard fuel mass penalty (kg).

The principle is similar for all time-adjustment calculations, and the time required to complete a sector is determined by adding all the calculated time adjustments to the proportion of the base lap time assigned to the particular sector in question.

Passing manoeuvres

Passing manoeuvres are modelled as follows: if a car enters a sector occupied by another car and the sector allows passing manoeuvres, the model determines whether the trailing car will be able to pass the leading car before the latter reaches the end of the sector. A successful passing manoeuvre depends on the relative pace and position of the two cars. If the leading car is too close to the end of the sector when the trailing car enters the sector (ie the gap between the cars is too large), a passing manoeuvre will not be allowed. If the gap is small enough and the relative pace of the trailing car is higher than that of the leading car, a passing manoeuvre will be allowed. The maximum gap size allowing passing manoeuvres may be determined empirically before the race for each race track. This principle is illustrated in Figure 2, where t_f denotes the time required by the trailing car to complete the sector, t_l denotes the time required by the leading car to complete the sector and t_g denotes the critical gap length to consider if a passing manoeuvre is to be attempted. A passing manoeuvre may only be attempted when $t_f - t_l < t_g$.

The value of t_g is the same in all sectors that allow passing manoeuvres. However, it may be refined and adjusted by assigning a unique value of t_g to every sector that allows passing manoeuvres. The average number of successful passing manoeuvres for a given race track is known in advance from observation of actual races and may be used to determine the value of t_g . If t_g is too small, too few passing manoeuvres will be observed during simulation, but if t_g is too large, too many passing manoeuvres will be observed.

Pit stops

Pit stops are modelled as follows: During a race a driver may temporarily suspend participation and visit his support team in the pits. The delay a driver experiences by performing a

pit stop comprises the time it takes to drive down the pit lane (ie from the pit lane entry to the pit lane exit, as illustrated in Figure 3) as well as the operational delay time. The operational delay time is determined by the fuel-pumping rate, the amount of fuel added during a pit stop and the efficiency of the pit-stop personnel.

In the simulation, the pit lane is also divided into a number of sectors. The start of the first pit lane sector coincides with the connection of two specific, consecutive sectors in the race track sector configuration, which is located at the pit lane entry. The end of the last sector in the pit lane then links with a different sector connection on the race track, which is located at the pit lane exit.

Model assumptions

The following model assumptions apply:

- The mixing of cars at the start of a race occurs only in the first sector of the first lap of a race.
- No car experiences the effects of air resistance during the first sector of the first lap of a race or in the pit lane (ie when performing a pit stop) due to the relatively low speeds reached in these sectors.
- No distinctions are made between the different sources that may cause cars to retire from a race. Input data to the simulation model are prepared based on this assumption, ie the model does not differentiate between failure types.
- All car retirements occur at the start of a sector.
- The performance of a tyre brand is reflected in the base lap time attribute of each car.

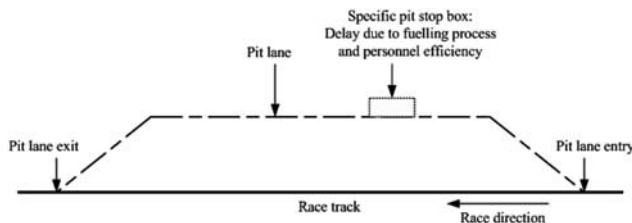


Figure 3 Illustration of a pit layout.

- The fuel consumption of each car remains constant for the duration of a race.
- The weather conditions remain dry and sunny for the duration of a race.
- Only one car may be overtaken at a time.

Model input data

Technical data for F1 cars are close-kept secrets. All teams guard their trade secrets in order to maintain any competitive advantage they might enjoy. However, some car performance data are freely available in the public domain, for example all lap times recorded by each driver during every practice session leading up to a race, data from previous races relating to pit-stop strategies, car failures and the mixing of cars at the start of a race. The model presented in this paper takes advantage of the availability of car performance data rather than relying on hard-to-come-by technical data. The fuel consumption and retirement probability for each driver can be determined from the freely available data. The race-starting fuel for each car can be estimated using a combination of historical data and various statistical techniques, that is if a specific driver recorded quick lap times throughout the practice sessions leading up to a race, but posted a slow qualifying time, without making any mistakes, it may be concluded that the driver carried a heavier fuel load compared to the other drivers.

Typical input data used in the model for characterizing a race car are shown in Table 1.

The mixing of cars at the start of a race is modelled using discrete empirical distributions. These distributions are constructed from historical performance data for each driver-and-associated-car combination. They give the probability of a driver gaining or losing a finite number of positions at the start of a race relative to the original starting position. The discrete empirical distributions of some drivers from the 2005 F1 season are shown in Figure 4.

The modelling of car mixing is best explained using an example. Assume the simulation model assigns a four-position gain, obtained from the associated discrete empirical distribution, to a driver who qualified in the third position.

Table 1 Typical race car characteristic input data for the 2005 Australian Grand Prix

Driver name	Car number	Starting position	Starting fuel (kg)	Base lap time (s)	Fuel consumption (kg/lap)	Probability of failure on any given lap (%)
Albers	1	17	50	93.144	2.93	0.54898
Freisacher	2	16	50	92.852	3.14	0.54898
Karthikeyan	3	12	56	87.970	2.8	0.32157
⋮	⋮	⋮	⋮	⋮	⋮	⋮
Barrichello	18	11	79	86.233	3.02	0.10910
Raikkonen	19	10	78	86.225	3	0.29174
Montoya	20	9	72	86.393	3	0.29174

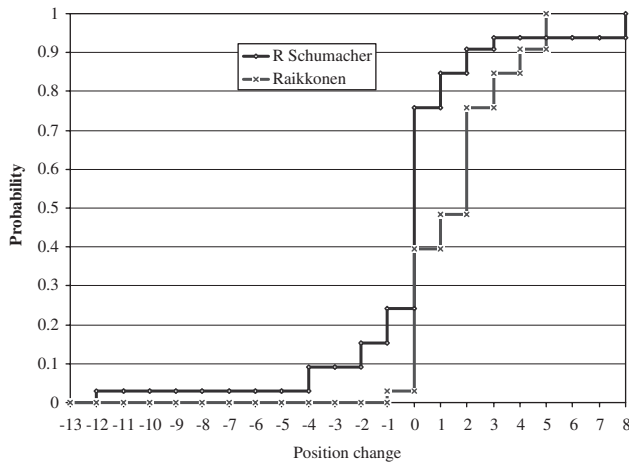


Figure 4 Typical discrete empirical distributions of start positions lost or gained (SP_{LG}).

Table 2 Typical race track characteristic input data for the Australian Grand Prix circuit

Sector number	Proportion of lap in sector (%)	Passing manoeuvres permitted?	Proportion of fuel consumption allocated to sector (%)
1	1.25	No	3.9
2	1.25	No	3.9
3	1.25	No	3.9
4	3.5	Yes	1.9
5	2.5	No	1.9
6	2.1	No	2.4
7	2.1	No	2.4
8	2.1	No	2.4
9	2.1	No	2.4
10	4.0	Yes	1.5
⋮	⋮	⋮	⋮

At the end of the first sector on the first lap of the race, the driver will emerge in the first position. Similarly, drivers starting from the back of the grid may only assume positions that are less than, or equal to the number of race entries (cars). Any position ties are broken arbitrarily.

Race track characteristic input data for the Australian Grand Prix circuit are shown in Table 2. As mentioned earlier, the values reflect the individual characteristics of a circuit. Passing manoeuvres occur only on certain parts (sectors) of a race track due to the nature of the specific track.

Pit-stop strategy data for all drivers are very important input parameters in the simulation model, that is, the number of pit stops scheduled, the corresponding lap number(s) on which the pit stop(s) will occur and the amount of fuel that will be added during each pit stop. Typical pit-stop strategies followed during the 2005 Australian Grand Prix are shown in Table 3. These values may be varied while using the proposed simulation model to predict the expected outcome of a race,

given a specific strategy. The required input data are summarized below:

1. Race car characteristic data

- The fuel quantity on board the car at the start of a race (kg).
- The position of the car on the starting grid.
- The discrete empirical distribution characterizing the car-mixing event at the start of a race.
- The fuel consumption (kg/lap).
- The base lap time (s).
- The retirement probability (%/lap).
- Scheduled pit-stop data
 - (i) The number of pit stops scheduled.
 - (ii) The corresponding lap number(s) on which each pit stop is scheduled to take place.
 - (iii) The amount of fuel that will be added during each scheduled pit stop (kg).

2. Race track characteristics data

- The number of laps in a race.
- The number of sectors in a lap.
- The standard fuel penalty, that is, the time adjustment calculated for every 10 kg of fuel on board a car (s/lap/10 kg).
- The gap size that permits passing manoeuvre attempts (s).
- The grid position penalty used to emulate the mixing of cars at the start of a race (cars at the front of the grid are able to get away sooner than cars further back).
- The proportion of a lap spent in each sector (%).
- The fuel consumption proportion assigned to each sector (%).
- The sector in which passing manoeuvre attempts are permitted.
- The probability of a successful passing manoeuvre (%).

3. General pit-stop data

- The fuel pump rate (kg/s).
- The time required to travel down the pit lane (s).
- The time required to enter and exit the pit-stop area (s).
- The distribution describing variations in the duration of a pit-stop (s).

Simulation model verification and validation

Since a simulation model is an abstraction of a subset of the real world, verification, validation and testing (VV&T) techniques are required to determine whether the model is an adequate representation of the real world it aims to imitate (Law and Kelton, 2000, p 265). There are more than 75 VV&T techniques (Banks, 1998, p 354) of which the informal

Table 3 Typical pit-stop strategy input data for some drivers in the 2005 Australian Grand Prix

Driver name	Car number	Number of pit stops planned	First stop		Second stop		Third stop		Fourth stop	
			Pit stop lap	Added fuel load (kg)	Pit stop lap	Added fuel load (kg)	Pit stop lap	Added fuel load (kg)	Pit stop lap	Added fuel load (kg)
Albers	1	2	16	59	35	70	–	–	–	–
Freisacher	2	2	15	63	34	79	–	–	–	–
Karthikeyan	3	2	19	59	39	56	–	–	–	–
Monteiro	4	2	18	58	36	70	–	–	–	–
Button	5	1	27	94	–	–	–	–	–	–
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

techniques are the most commonly used. These include auditing, reviews, inspection and face validation.

The model was verified and validated using a spectrum of Grands Prix and their respective race tracks, each allowing a different average speed. The following three race tracks were selected to give a mix of tracks with different straight/corner combinations:

1. *The Australian Grand Prix*: The Albert Park race track in Melbourne has a combination of high-speed straights and low- to medium-speed corners—a medium-speed race track.
2. *The Hungarian Grand Prix*: The Hungaroring race track in Budapest consists predominantly of sharp low- to medium-speed corners—a low-speed race track.
3. *The Italian Grand Prix*: The Autodromo Nazionale di Monza race track near Monza consists primarily of long, high-speed straights—a high-speed race track.

The following model output parameters are compared to the associated real-world values of the 2005 Australian, Hungarian and Italian Grands Prix:

1. The positions of the cars at the end of a race.
2. The race times at the end of a race (the time it took each driver to complete the race).
3. The number of successful passing manoeuvres during a race.

The Spearman rank correlation coefficient (Walpole and Myers, 1993) is used to compare the positions of the cars at the end of each race. The null hypothesis, denoted by H_0 , states that there is no correlation between actual and simulated data sets, that is, the ranking of the values (car positions) in the sets are random. The following hypothesis is tested at a significance level of $\alpha = 5\%$:

$$H_0: \rho = 0$$

$$H_1: \rho \neq 0$$

The rank correlation coefficient is given by

$$r_s = 1 - \frac{6 \sum_{i=1}^m d_i^2}{m(m^2 - 1)} \quad (2)$$

where d_i denotes the difference between the ranks assigned to the i th pair of observations in the two data sets and m denotes the number of data pairs. In Table 4, actual positions of the 2005 Australian, Hungarian and Italian Grands Prix are shown with the corresponding simulated results. The columns labelled 'Estimated' represent the average car positions obtained from simulating 50 independent replications as prescribed by Law and Kelton (2000, pp 513–514).

The null hypothesis is rejected in all three cases, which implies that the rankings in each of the three cases are not independent. Since in all cases the Spearman rank correlations are very close to 1, we conclude that the simulation model estimates the rankings satisfactorily.

In Table 5, race time (seconds) classifications of the 2005 Australian, Hungarian and Italian Grands Prix are shown with the corresponding simulated results. The columns labelled 'Estimated' represent the average race times obtained from simulating 50 independent replications as prescribed by Law and Kelton (2000, pp 513–514). Since the observations are paired, and the respective sets of observations are not normally distributed, the Wilcoxon matched-pairs test is applied using Statistica (StatSoft, accessed 2008) to test if the two samples (actual race times and simulated race times) differ significantly. The test assumes that the differences of matched pairs are centred around a common median $\theta_1 = \theta_2 = \theta_3$, and the following hypothesis is tested at $\alpha = 5\%$:

$$H_0: \theta_1 = \theta_2$$

$$H_1: \theta_1 \neq \theta_2$$

Since H_0 cannot be rejected, it implies that there is insufficient evidence that the actual race times differ from the race times as reported by the simulation model.

In Table 6, the number of successful passing manoeuvres during the selected races are compared to the corresponding simulated results. The expected number of passing manoeuvres is again based on 50 simulation replications (Law and

Table 4 Validation of the positions at the end of the 2005 Australian, Hungarian and Italian Grands Prix

Driver name	Car number	Australian Grand Prix		Hungarian Grand Prix		Italian Grand Prix	
		Actual	Estimated	Actual	Estimated	Actual	Estimated
Albers	1	20	20.00	15	15.00	19	18.3
Freisacher	2	19	17.00	18	18.00	18	20
Karthikeyan	3	15	14.74	12	12.00	20	18.7
Monteiro	4	16	16.00	13	13.00	17	16.04
Button	5	11	10.38	5	4.86	8	8.04
Heidfeld	6	19	19.00	6	8.94	7	6.62
Webber	7	5	5.80	7	10.50	14	15.42
Klien	8	7	5.44	19	19.00	13	11.14
Trulli	9	9	8.80	4	3.54	5	4.06
Coulthard	10	4	4.36	20	20.00	15	15.6
Villeneuve	11	13	13.30	16	16.00	11	10.4
Fisichella	12	1	1.00	9	7.40	3	4.34
Massa	13	10	10.54	14	14.00	9	10.38
M Schumacher	14	18	18.00	2	1.92	10	11.44
Sato	15	14	13.02	8	6.14	16	13.42
R Schumacher	16	12	11.92	3	3.42	6	4.26
Alonso	17	3	7.62	11	9.04	2	1.98
Barrichello	18	2	2.06	10	8.92	12	9.42
Raikkonen	19	8	8.06	1	1.32	4	5.72
Montoya	20	6	2.96	17	17.00	1	1.02
<i>P</i> -value		< 0.05		< 0.05		< 0.05	
r_s		0.972		0.970		0.970	
Decision		Reject H_0		Reject H_0		Reject H_0	

Table 5 Validation of the race times (seconds) at the end of the 2005 Australian, Hungarian and Italian Grands Prix

Driver	Car number	Australian Grand Prix		Hungarian Grand Prix		Italian Grand Prix	
		Actual	Estimated	Actual	Estimated	Actual	Estimated
Albers	1	1417.256*	1420.285	5915.142	6034.477	4526.715	4504.485
Freisacher	2	5053.320	5056.067	2302.411*	2168.084	4493.706	4511.013
Karthikeyan	3	5055.224	5064.739	5890.414	5908.671	4545.892	4505.544
Monteiro	4	5094.584	5108.650	5886.357	5885.477	4482.264	4509.839
Button	5	5066.004	5076.205	5904.384	5909.102	4532.294	4471.261
Heidfeld	6	3675.221*	3664.277	5913.927	5894.790	4513.302	4515.698
Webber	7	5061.744	5069.808	5853.302	5906.567	4475.624	4488.562
Klien	8	5072.882	5069.668	0.000*	178.974	4474.359	4494.007
Trulli	9	5081.201	5098.154	5899.773	5888.603	4502.445	4503.150
Coulthard	10	5079.299	5065.774	0.000*	91.970	4491.151	4511.030
Villeneuve	11	5050.071	5057.525	4871.605*	4803.257	4469.901	4487.328
Fisichella	12	5029.673	5031.312	5887.300	5882.873	4486.634	4507.268
Massa	13	5079.625	5099.598	5918.861	5895.330	4544.072	4491.852
M Schumacher	14	3703.104*	3682.629	5881.133	5866.156	4564.729	4490.035
Sato	15	5071.446	5059.028	5866.472	5876.309	4492.104	4500.709
R Schumacher	16	5073.913	5051.452	5881.681	5876.577	4512.584	4509.705
Alonso	17	5051.297	5082.854	5895.467	5891.845	4471.138	4457.084
Barrichello	18	5067.945	5044.472	5894.241	5888.335	4473.57	4502.638
Raikkonen	19	5061.081	5079.545	5845.552	5848.993	4491.434	4509.480
Montoya	20	5067.861	5052.211	3448.592*	3393.244	4468.659	4447.814
<i>P</i> -value		0.881		0.709		0.823	
Decision		Do not reject H_0		Do not reject H_0		Do not reject H_0	

*Cars that did not finish the race. Note that these cars still receive final positions, as shown in Table 4.

Table 6 Validation of the number of successful passing manoeuvres during the 2005 Australian, Hungarian and Italian Grands Prix

<i>Australian Grand Prix</i>			<i>Hungarian Grand Prix</i>			<i>Italian Grand Prix</i>		
<i>Actual</i>	<i>Estimated</i>	<i>Deviation</i>	<i>Actual</i>	<i>Estimated</i>	<i>Deviation</i>	<i>Actual</i>	<i>Estimated</i>	<i>Deviation</i>
4	4.12	+2.91%	11	11.02	+0.18%	9	8.92	−0.90%

Table 7 Final positions due to different pit-stop strategies for Trulli (Car 9) and R Schumacher (Car 16)—2005 Australian Grand Prix

Actual race strategy		Simulated race strategies		
		Scenario 1	Scenario 2	Scenario 3
Car 9				
Number of planned pit stops	2	3	2	2
Lap numbers of planned pit stops	18, 36	18, 32, 50	18, 46	18, 46
Position	9	8.10	7.44	7.70
Car 16				
Number of planned pit stops	2	2	1	2
Lap numbers of planned pit stops	19, 40	19, 40	19	19, 46
Position	12	12.08	13.60	9.36
Sum of positions	21	20.18	21.04	17.06

Table 8 Final positions due to different pit-stop strategies for Trulli (Car 9) and R Schumacher (Car 16)—2005 Hungarian Grand Prix

<i>Actual race strategy</i>		<i>Simulated race strategies</i>		
		<i>Scenario 1</i>	<i>Scenario 2</i>	<i>Scenario 3</i>
<i>Car 9</i>				
Number of planned pit stops	3	2	3	2
Lap numbers of planned pit stops	13, 33, 52	13, 45	13, 33, 52	13, 35
Position	4	4.58	3.20	3.92
<i>Car 16</i>				
Number of planned pit stops	3	2	2	2
Lap numbers of planned pit stops	15, 35, 54	15, 47	15, 50	15, 47
Position	3	2.76	4.20	2.14
Sum of positions	7	7.34	7.40	6.06

Kelton, 2000, pp 513–514). As these are single-point estimations, face validation (Banks, 1998) is used to compare the actual and simulated values. The percentage deviations are considered small and it is accepted that the passing manoeuvres are modelled correctly.

The validation analyses demonstrate that, for the given output parameters, no significant differences exist between the real-world observations and those reported by the simulation model. The model is therefore considered an adequate representation of a real-world F1 race.

Simulation results

Before a specific race, the qualifying lap times of each car can be applied to the simulation model and a team's pit-stop strategy can be planned for that race. Planners can use their expertise to decide on the number of pit-stops and

the lap numbers on which these should occur. A chosen scenario cannot guarantee success, but the team can follow the suggested strategy knowing that it is the expected best of the scenarios considered.

Tables 7–9 compare actual and simulated race results obtained from simulating alternative race strategies for team Toyota during the 2005 Australian, Hungarian and Italian Grand Prix, respectively. The columns labelled 'Scenario 3' indicate the best simulated race strategy for the team, because in general the expected position for *both* cars improved. The strategies that favoured only one car in the team are unacceptable since the objective is to improve team performance.

The results show that the specific team could have done better had a different pit-stop strategy been followed. The model can thus be used to plan strategies for future events but a database describing other race tracks must first be compiled.

Table 9 Final positions due to different pit-stop strategies for Trulli (Car 9) and R Schumacher (Car 16)—2005 Italian Grand Prix

<i>Actual race strategy</i>		<i>Simulated race strategies</i>		
		<i>Scenario 1</i>	<i>Scenario 2</i>	<i>Scenario 3</i>
<i>Car 9</i>				
Number of planned pit stops	2	2	2	3
Lap numbers of planned pit stops	20, 40	20, 40	20, 40	20, 28, 41
Position	5	3.44	3.36	3.56
<i>Car 16</i>				
Number of planned pit stops	2	2	3	2
Lap numbers of planned pit stops	22, 41	22, 35	22, 30, 38	22, 41
Position	6	4.92	6.14	4.62
Sum of positions	11	8.36	9.50	8.18

Conclusions

A discrete-event simulation model capable of simulating F1 races was shown to be possible as well as practical, and the time-adjustment approach developed in particular allows racing events to be modelled on an aggregate basis, thus avoiding the need to model each and every intricacy of a race car. The model only requires input data that are freely available in the public domain. We validated the model and successfully improved on the race strategies employed by team Toyota during the 2005 Australian, Hungarian and Italian Grands Prix. This provided evidence to support the usefulness of the model.

The planning and analysis of race strategies using the model could assist the decision-maker(s) of a racing team by providing them with the best possible opportunity to maximize the performance of their resources, namely to score as many points as possible during a season. In this multi-million dollar sport such an advantage is invaluable.

Future work includes refining the model to incorporate changing weather conditions as well as the effect of major accidents. Teams in F1 are increasingly relying on advanced simulation technologies in their search for a competitive advantage. The ultimate goal is thus to develop a model capable of analysing race strategies in real time. The decision-maker(s) will subsequently be able to react with confidence to racing incidents within seconds from the time they occur.

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